25901043 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCFTA102 - MATHEMATICAL AND STATISTICAL METHODS IN FOOD TECHNOLOGY

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or two sentence

Answer any **5** questions

Each question carries **2** marks

1. If $A_{=}\begin{pmatrix}1 & 3\\ 2 & 4\end{pmatrix}$, Find its determinant? [U] / [CO1]
2. Define Nil potent matrix [K] / [CO1]
3. Differentiate $(x\sin x)^3$ [U] / [CO2]
4. Find the derivative of $6x^{-2}+7x^3-3x^2+8x+98$? [U] / [CO2]
5. Write any two merits of standard deviation. [K] / [CO3]
6. What is coefficient of variation? [K] / [CO3]
7. Define Chi -Square distribution. [K] / [CO4]
8. Give an example of Standard Error. [K] / [CO4]

[2x5 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **6** marks

9. Compute the Invers $A_{=}\begin{pmatrix}2 & -1\\ 1 & 3\end{pmatrix}$ [U] / [CO1]

10. Explain triangular matrix with example [U] / [CO1]

11. Evaluate $\int x(x+1)(x+2)\mathrm{d}x$ & integrate $(3x^2+2x-3)/x^3$ [U] / [CO2]

12. Integrate : $x^3 \log x$ by parts. [U] / [CO2]

13. What are the properties of a good measure of dispersion? [U] / [CO3]

14. Calculate Arithmetic Mean of the daily income of 10 families given below [K] / [CO3]

    Income : 18, 20, 32, 35, 40, 55, 70, 80, 82, 90

15. Write a Short note on Z- distribution [U] / [CO4]

16. Find P(Z>1.8) ? [U] / [CO4]

[6x5 = 30]

Part C

Essay

Answer any **2** questions

Each question carries **15** marks

17. If $A_{=}\begin{pmatrix}-4 & -3 & -3\\ 1 & 0 & 1\\ 2 & -2 & 1\end{pmatrix}$, Show that adjA = A [U] / [CO1]

18. Differentiate $(x^3+7)(3x^2-5)(x+1)$ [U] / [CO2]

19. FindQ3, D2 and P40 for the following distribution [U] / [CO3]

Marks :	*10 - 15*	*15 - 20*	*20 - 25*	*25 - 30*	*30 - 35*
No, of students :	*2*	*7*	*18*	*10*	*3*

20. Explain the properties of F - distribution. [U] / [CO4]

[15x2 = 30]